

TIAGo Pro: Joint Limits Reference

From tiago_pro.urdf.xacro. Hard URDF limit values (tighter safety_controller soft limits apply during normal ros2_control operation).

Section	Joint	Type	Min	Max	Effort	Velocity
Torso	torso_lift_joint	Prismatic	−0.001 m	0.35 m (35 cm)	2000 N	0.035 m/s
Head	head_1_joint (pan)	Revolute	−1.309 rad (−75.0°)	1.309 rad (75.0°)	5.197 Nm	3.0 rad/s
	head_2_joint (tilt)	Revolute	−1.047 rad (−60.0°)	0.349 rad (20.0°)	2.77 Nm	3.0 rad/s
Left arm	arm_left_1_joint	Revolute	−0.524 rad (−30.0°)	4.712 rad (270.0°)	43.0 Nm	1.95 rad/s
	arm_left_2_joint	Revolute	−2.443 rad (−140.0°)	1.134 rad (65.0°)	43.0 Nm	1.95 rad/s
	arm_left_3_joint	Revolute	−2.618 rad (−150.0°)	2.618 rad (150.0°)	26.0 Nm	3.95 rad/s
	arm_left_4_joint	Revolute	−2.443 rad (−140.0°)	1.134 rad (65.0°)	26.0 Nm	3.95 rad/s
	arm_left_5_joint	Revolute	−3.665 rad (−210.0°)	1.571 rad (90.0°)	26.0 Nm	3.95 rad/s
	arm_left_6_joint	Revolute	−1.885 rad (−108.0°)	3.002 rad (172.0°)	26.0 Nm	3.95 rad/s
	arm_left_7_joint	Revolute	−2.443 rad (−140.0°)	2.443 rad (140.0°)	26.0 Nm	3.95 rad/s
Right arm	arm_right_1_joint	Revolute	−4.712 rad (−270.0°)	0.524 rad (30.0°)	43.0 Nm	1.95 rad/s
	arm_right_2_joint	Revolute	−2.443 rad (−140.0°)	1.134 rad (65.0°)	43.0 Nm	1.95 rad/s
	arm_right_3_joint	Revolute	−2.618 rad (−150.0°)	2.618 rad (150.0°)	26.0 Nm	3.95 rad/s
	arm_right_4_joint	Revolute	−2.443 rad (−140.0°)	1.134 rad (65.0°)	26.0 Nm	3.95 rad/s
	arm_right_5_joint	Revolute	−1.571 rad (−90.0°)	3.665 rad (210.0°)	26.0 Nm	3.95 rad/s
	arm_right_6_joint	Revolute	−1.885 rad (−108.0°)	3.002 rad (172.0°)	26.0 Nm	3.95 rad/s
	arm_right_7_joint	Revolute	−2.443 rad (−140.0°)	2.443 rad (140.0°)	26.0 Nm	3.95 rad/s
Gripper (driven)	gripper_left_finger_joint	Prismatic	−0.001 m	0.07 m (7 cm)	10.0 N	2 m/s
	gripper_right_finger_joint	Prismatic	−0.001 m	0.07 m (7 cm)	10.0 N	2 m/s
Gripper (mimic)	inner finger, L/R	Revolute	−42.8°	8.0°	0.1 Nm	2 rad/s
	outer finger, L/R	Revolute	−42.8°	8.0°	0.1 Nm	2 rad/s
	fingertip, L/R	Revolute	−8.0°	42.8°	0.1 Nm	2 rad/s
	finger screw, right	Prismatic	−0.001 m	0.07 m	10.0 N	2 m/s
Wheels	wheel_front_right_joint	Continuous	no limit	no limit	6.0 Nm	100.0 rad/s
	wheel_front_left_joint	Continuous	no limit	no limit	6.0 Nm	100.0 rad/s
	wheel_rear_right_joint	Continuous	no limit	no limit	6.0 Nm	100.0 rad/s
	wheel_rear_left_joint	Continuous	no limit	no limit	6.0 Nm	100.0 rad/s

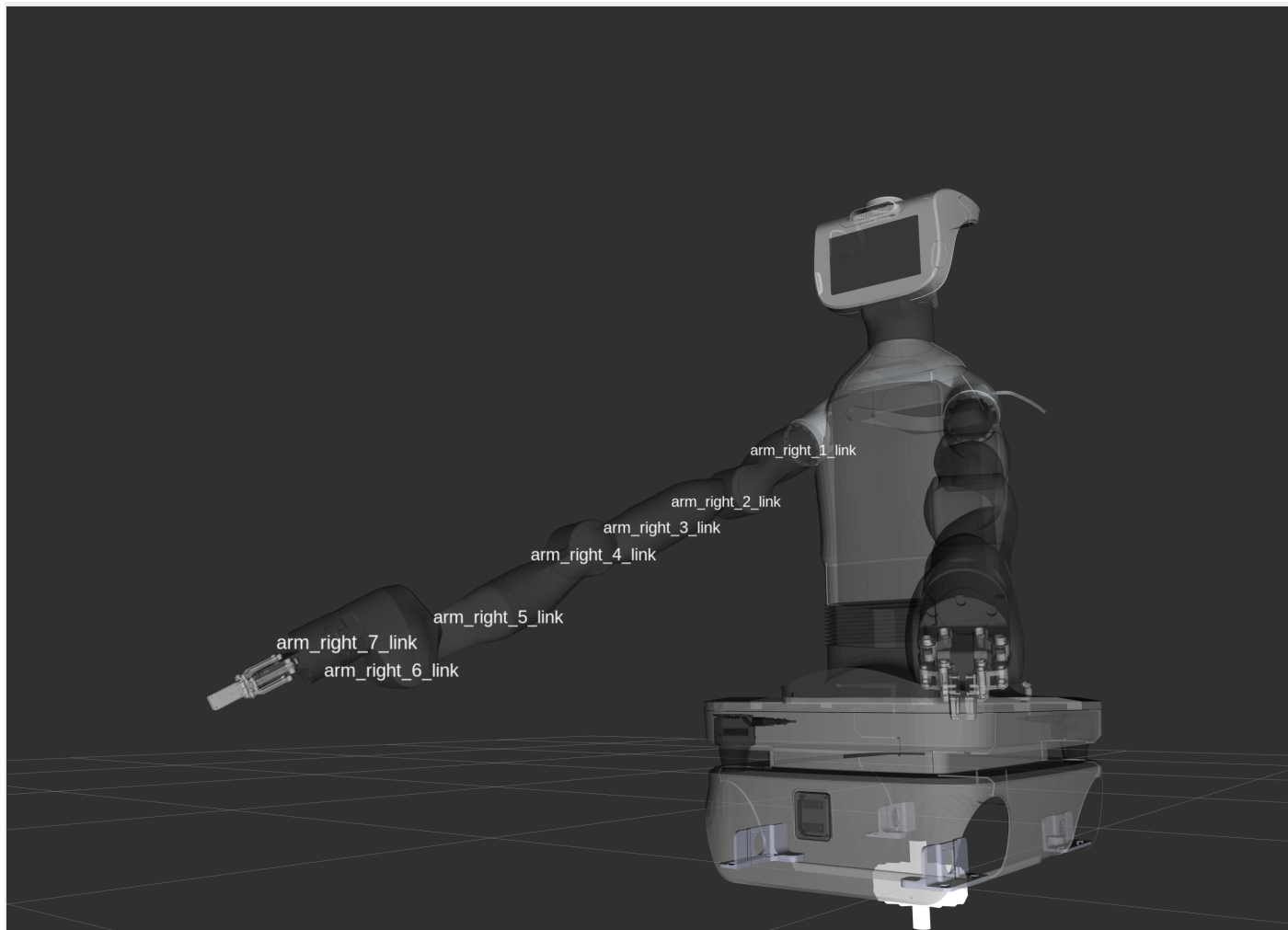
Only *_finger_joint per gripper is commanded directly; mimic joints follow it automatically. Wheels use a mecanum drive; whole-body cmd_vel is separately capped at −1.0 to 1.0 m/s linear and −1.0 to 1.0 m/s² acceleration.

Sensor Specifications

From Gazebo sensor plugins in tiago_pro.urdf.xacro.

Sensor	Parameter	Value
Head RGB camera	Topic	/head_front_camera/color
	Resolution	640 × 360 px
	Horizontal FOV	1.518 rad (87.0°)
	Vertical FOV	0.977 rad (56.0°)
	Update rate	30 Hz

Sensor	Parameter	Value
Head RGBD depth camera	Topic	/head_front_camera/depth
	Resolution	640 × 360 px
	Horizontal FOV	1.518 rad (87.0°)
	Vertical FOV	0.977 rad (56.0°)
	Depth clip (near)	0.2 m
	Depth clip (far)	8.0 m
	Update rate	30 Hz
Front LiDAR	Topic	/scan_front_raw
	Type	gpu_lidar (SICK TIM551)
	FOV	−2.339 to 2.356 rad (−270°)
	Range	0.05 – 25.0 m
	Samples	818 (0.33°/step)
	Noise	Gaussian, $\sigma = 0.01$
	Update rate	10 Hz
Rear LiDAR	Topic	/scan_rear_raw
	Type	gpu_lidar (SICK TIM551)
	FOV	−2.339 to 2.356 rad (−270°)
	Range	0.05 – 25.0 m
	Samples	818 (0.33°/step)
	Noise	Gaussian, $\sigma = 0.01$
	Update rate	10 Hz
Base IMU	Topic	/base_imu
	Update rate	100 Hz
	Rate noise (σ)	2×10^{-4} rad/s
	Accel noise (σ)	1.7×10^{-2} m/s ²



Visualization of arm joints in RViz